

Dynamic Analysis Instrument Element Procedure

Revision

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SME

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Abstract

This document describes the procedure used to perform the instrument element activity described in the **AVCDL** ^[1] secondary document **Dynamic Analysis Report** ^[2].

Group / Owner

Security / Security Architect

Motivation

This document is motivated by the need to have runtime-specific, cybersecurity-related feedback in the development of software for use within safety-critical, cyber-physical systems.

Note: Within the context of this document, the terms ***security*** and ***cybersecurity*** are used interchangeably. It is presumed that the term ***security*** is being used in reference to ***cybersecurity*** and not ***physical security***.

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Audience

The audience of this document is the cybersecurity practitioner who will be conducting cybersecurity-specific dynamic analysis.

Completeness of Output

Since dynamic analysis is an as-is analysis, it is critical to ensure that the information gathered is as accurate and complete as possible. As with any cybersecurity assessment, it is not the place of the cybersecurity SME to make assumptions on the part of engineering. All information should be sourced from and confirmed by the owner of the element under consideration.

Disposition of Output

This activity does not produce documentation as an output.

Entry Criteria

This document assumes that the reader understands the purpose of cybersecurity-specific dynamic analysis. Further, that the reader has read and understood the AVCDL **Dynamic Analysis Report** ^[2] secondary document.

Prerequisites – Security SME

Qualifications

It is required that the threat modeling SME is both a qualified and trained security architect (shown above on title page as **Owner**) as defined by the **NIST NCWF** role SP-ARC-002 and detailed in the **NCWF Roles - Security Architect** section of the **AVCDL Background Material** elaboration document ^[4].

Knowledge

It is required that the cybersecurity SME understands the purpose of cybersecurity-specific dynamic analysis.

Background Information

It is required that the cybersecurity SME has read and understands the AVCDL **Dynamic Analysis Report** secondary document. Additionally, that the cybersecurity SME has taken training relevant to this activity.

Prerequisites – DevOps SME

Qualifications

It is required that the DevOps SME is both a qualified and trained DevOps SME with expertise in the build system used for the element under consideration in the context of the dynamic analysis tool being employed.

Note: The specific qualifications and training of DevOps SMEs necessary to assert expertise in providing accurate information regarding the build system used for the element under consideration is outside the scope of this document.

Note: The DevOps group is responsible for determining and making available the appropriate SME for the modifications to build system needed for the tool under consideration.

Knowledge

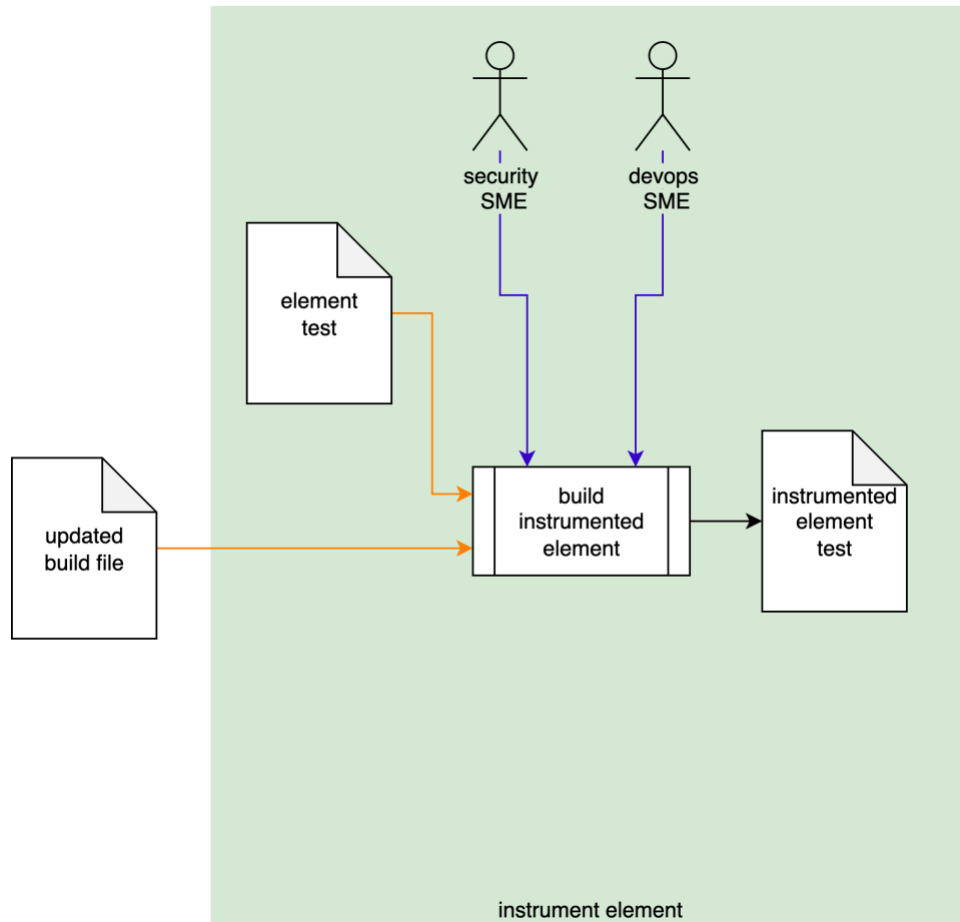
It is required that the DevOps SME have expertise in the build system used for the element under consideration in the context of the dynamic analysis tool being employed.

Prerequisites – Input Materials

It is required that the cybersecurity SME acquire all relevant documentation related to the tool under consideration necessary to complete the procedure.

Instrument Element Activity

The workflow diagram for the **instrument element** activity of the **Dynamic Analysis** process is shown below.



Using the **Updated Build File** created in the **Dynamic Analysis - Establish Settings** activity [\[3\]](#), the Security and DevOps SMEs will take the **Element Test** and make adjustments to allow for use in a dynamic analysis context. An **Instrumented Element Test** will be produced.

Note: In some cases, the Security SME may perform this activity alone or the activity may even be fully automated.

Note: The scope of the **Element Test** may be as small as a single file, or as large as the entire project.

Methodology

Because of the diversity of individual dynamic analysis tool, the implementation of this activity is deferred to the documentation provided by the chosen dynamic analysis tool.

Considerations

Since dynamic analysis is a runtime activity typically utilized via a test suite, there are several things that need to be taken into consideration.

Termination on Error

Most dynamic analysis tools default to a termination-on-failure behavior. It is therefore important to adjust the test suite to take this into consideration.

Issue Occlusion

Discovered issues may mask more fundamental ones. Be sure to adjust the test to verify that the issue discovered is the root issue.

Footprint (Code / Memory)

The instrumentation of code for dynamic analysis purposes may incur non-trivial executable size and memory footprint impacts. These may impact both size and execution timing.

Cybersecurity Side-effects

An important consideration is that because dynamic analysis require intrusive access to data structures within the code, there may be a significant impact on other cybersecurity-related tooling. These should probably be disabled in order to not trigger false positives in those tools.

Exit Criteria

This procedure is considered complete once the **Instrumented Element Test** has been created.

References

1. **AVCDL** (AVCDL primary document)
2. **Dynamic Analysis Report** (AVCDL secondary document)
3. **Dynamic Analysis Element Selection Procedure** (AVCDL tertiary document)
4. **AVCDL Background Material** (AVCDL elaboration document)